



Shirpur Education Society's
R. C. PATEL INSTITUTE OF TECHNOLOGY, SHIRPUR
An Autonomous Institute

[Affiliated to Dr. Babasaheb Ambedkar Technological University, Lonere]



आर. सी. पटेल इन्स्टिट्यूट ऑफ टेक्नॉलॉजी, शिरपूर
(स्वायत्त महाविद्यालय)

Programme: B.Tech in Computer Engineering

Year: III/Semester VI (Exam Year: 2023-2024)

Subject: Information Security

Date: 29 Jul 2024

Time: 09:00 am - 12:00 pm (03:00 Hrs.)

Max Marks: 75

RE END SEMESTER EXAMINATION EVEN SEM VI (2023-2024)

Instructions: 1. This question paper contains 2 pages

2. Answer to each new question to be started on a fresh page.

3. Figure in right hand side indicates full marks

4. Assume suitable data wherever necessary and clearly mention the same.

1. 15
 - a. 8
 - i. Describe the encryption process of a Simple Columnar Transposition Cipher. Include the steps involved and perform encryption using the following information: 8
PLAINTEXT: THE QUICK BROWN FOX JUMPS OVER THE LAZY DOG , **KEY:** COLUMN (IGNORE SPACES)
----- OR -----
 - ii. Write down the steps to convert the plaintext into cipher text using Playfair Cipher. 8
Also Derive the Ciphertext from the following plaintext using Playfair cipher:
Plaintext = instrument KEY = monarchy
 - b. **List Block cipher modes of operation and describe any 2.** 7
2. 15
 - a. 7
 - i. **Discuss about possible attacks on RSA Cryptosystem. If $p = 13$, $q = 5$ and $e = 7$, Evaluate the value of d and cipher text if plaintext = 6. Make sure e and d are not same values.** 7
----- OR -----
 - ii. What is Man in the middle attack how it is possible in Diffie Hellman Algorithm? 7
If Alice and Bob have chosen prime value $q = 13$ and primitive root = 2. If Alice's secret key is 5 and Bob's secret key is 4, what is the secret key they exchanged?
 - b. **Explain Double DES algorithm. What were the drawbacks of Double DES (2-DES)?** 8
3. 15
 - a. Explain the basic working principle of the SHA-1 (Secure Hash Algorithm 1) and how it differs from the MD5 algorithm in terms of security and collision resistance. 10



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(स्वायत्त महाविद्यालय)

Programme: B.Tech in Computer Science and Engineering (Data Science)

Year: III/Semester V(Exam Year: 2023-2024)

Subject: Information Security

Max. Marks: 75

Date: 04 Jan 2024

Time: 09:30 am to 12:30 pm (03:00 Hrs.)

END SEMESTER EXAMINATION ODD SEM-V(Acad. Year:2023-2024)

Instructions:

1. This question paper contains 2 pages.
2. Answer to each new question to be started on a fresh page.
3. Figure in right hand side indicates full marks
4. All five questions are compulsory.
5. Assume suitable data wherever required, but justify it.

Question No.		Max. Marks
Q1 (a)	Enumerate and explain three common types of cyber-attacks, highlighting the potential impact on the confidentiality, integrity, and availability of data.	[05]
	OR	
Q1 (b)	"Overhearing of signals by using the conductor" falls under what category of threat and performed at which layer of OSI. Justify.	[05]
Q1 (c)	Describe the key components of the TCP/IP protocol suite. Explain the roles of the major protocols in the suite and how they facilitate communication across networks.	[10]
Q2 (a)	Explain the key steps involved in the Data Encryption Standard (DES) encryption process. Include details about the initial permutation, key generation, the Feistel network structure. Discuss any vulnerabilities or limitations associated with DES.	[08]
	OR	
Q2 (b)	Explore the security considerations associated with block cipher modes of operation. Discuss the concept of initialization vectors (IVs) and their role in enhancing security.	[08]
Q2 (c)	i)Apply the Extended Euclidean Algorithm to find the coefficients x and y in the equation $ax + by = \gcd(a, b)$, where $a=17$ and $b=26$. II) Using the results from part 1, determine the modular multiplicative inverse of 17 modulo 26. Explain the steps involved in the process.	[07]
Q3 (a)	Assume you have chosen $p=17$ and $q=19$ as prime numbers for an RSA system. Explain RSA and Calculate public key and private key pairs. Then, encrypt the message $M=10$.	[05]
	OR	
Q3 (b)	Alice and Bob want to exchange a secret key using Diffie-Hellman. An attacker, Mallory, intercepts their communication. Alice chooses $p=17$ and $g=3$. Bob chooses the same p and g . Mallory intercepts Alice's public key ($A=5$) and Bob's public key ($B=8$), replaces them with her own public key ($M=13$), and establishes separate shared keys with Alice and Bob. Calculate the shared keys between Mallory and Alice (K_{MA}) and between Mallory and Bob (K_{MB}).	[05]
Q3 (c)	Two parties want to establish the secured communication channel with following security goals give corresponding security solution: 1. Confidentiality 2. Confidentiality and Integrity 3. Confidentiality, Integrity and Non-repudiation	[10]

Q3 (d)	OR Provide a comprehensive explanation of the SHA-1 algorithm, emphasizing its structure, key steps, and cryptographic properties.	[10]
Q4 (a)	Discuss the advantages and challenges associated with biometric authentication methods, highlighting the importance of factors such as accuracy, uniqueness, and non-repudiation.	[08]
Q4 (b)	OR Outline the key steps involved in a typical Kerberos authentication process, including the roles of the Key Distribution Center (KDC), Ticket Granting Ticket (TGT), and Service Tickets.	[08]
Q4 (c)	Differentiate between Signature-based and Heuristic-based detection methods employed by IDS. Highlight the advantages and limitations of each type	[07]
Q5 (a)	Provide examples of common network attacks that involve spoofing and explain how attackers utilize spoofing to compromise network integrity.	[05]
Q5 (b)	OR Define Denial of Service (DoS) attacks and briefly explain the following types: Syn flooding, and Echo charge	[05]
Q5 (c)	Compare the advantages and disadvantages of Tunnel Mode and Transport Mode in IPsec.	[10]

All the Best!



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 (स्वायत्त महाविद्यालय)

Programme: B.Tech in Computer Science and Engineering (Data Science)

Year: IV/Semester VII (Exam Year: 2025-2026)

Subject: Information Security

Max Marks: 65

Date: 04 Dec 2025

Time: 09:30 am - 11:30 am (02:00 Hrs.)

END SEMESTER EXAMINATION ODD SEM-VII-2025-REGULAR

Instructions:

1. This question paper contains 2 pages
2. Answer to each new question to be started on a fresh page.
3. Figure in right hand side indicates full marks
4. All four questions are compulsory.
5. Assume suitable data wherever required, but justify it.
6. Support your answers with neat, labelled diagrams, wherever necessary.

- | | | |
|-----------------------|--|----|
| 1. | | 20 |
| A. | Explain the different security goals, and describe the security services used to achieve these goals. | 5 |
| B. | What is Fermat's little theorem? Given two integers $a=168$ and $b=32$, use the Extended Euclidean algorithm to: | 7 |
| | 1. Find the greatest common divisor (GCD) of a and b . | |
| | 2. Find integers s and t such that $s * a + t * b = \text{GCD}(a, b)$ | |
| | Show all the steps clearly and verify your answer by plugging back the values into the equation. | |
| C. .. | | 8 |
| 1. | How can confidentiality be achieved using Public Key cryptography? An RSA system uses primes $p = 11$ and $q = 37$. | 8 |
| 1. | If the public exponent $e = 7$, what is the value of the private exponent d ? Show all steps using the Extended Euclidean Algorithm. | |
| 2. | What is the cipher text for plaintext $m = 20$? | |
| | Elaborate using the steps followed in the Fast Exponent algorithm. | |
| ----- OR ----- | | |
| 2. | How the key exchange is performed using the Diffie-Hellman algorithm? discuss its vulnerability to a Man-in-the-Middle attack. Given that Alice and Bob use Diffie-Hellman with prime $p=31$ and base $g=5$, where Alice chooses private key $a=6$ and Bob chooses private key $b=16$: | 8 |

1. What is the shared secret key established between Alice and Bob?
 2. How can an intruder intercept and manipulate the key exchange?
 3. What are the intercepted values and the shared secrets established by the intruder with Alice and Bob respectively?
2. 15
- A. Compare and analyze the security challenges encountered in key distribution using KDC versus Public Key Certificates. 5
- B. .. 10
1. Given a 64-bit input to the DES analyse how S-boxes and expansion box performs the operation. Analyze why DES is susceptible to differential cryptanalysis attack. 10
- **OR** -----
2. Analyze the differences between ECB and CBC modes in block cipher operation. For a given plaintext block, how the encryption process differs in each mode and discuss the security implications? 10
3. 15
- A. Analyse the components and role of Public Key Infrastructure (PKI) in securing network communications. Explain how PKI supports authentication and confidentiality. 5
- B. .. 10
1. In a given scenario where parties A and B want to securely exchange keys and data over an insecure channel, design a secure communication setup. Analyse the steps involved using authentication, key distribution protocols (KDC, Needham-Schroeder). 10
- **OR** -----
2. Analyse and discuss a scenario where mutual authentication is required between two services, and how Kerberos supports this via tickets and session keys. 10
4. 15
- A. What is DoS attack? Explain different types of DoS attacks. Discuss the detection and prevention strategies to employ against DoS attack. 5
- B. .. 10
1. Analyse the role of IPsec in securing IP communications, examining its key security features with comparative analysis between Tunnel mode and Transport mode. 10
- **OR** -----
2. Discuss the major functions of an Intrusion Detection system, (IDS). Explain the different detection techniques which IDS employs to check network packets. 10

SVKM
D. J. Sanghvi College of Engineering
Programme: B.Tech in Computer Engineering
Year: III/Semester VI (Exam Year: 2024-2025)

Subject: Information Security

Date: 24 May 2025

Time: 09:00 am - 11:00 am (02:00 Hrs.)

Max Marks: 65

FINAL EXAMINATION(2024-2025)

Instructions:

1. This question paper contains 3 pages
2. Answer to each new question to be started on a fresh page.
3. Figure in right hand side indicates full marks
4. Draw Neatly labelled Diagrams wherever applicable.
5. Assume Suitable data wherever required.

- | | | |
|------------------------------|---|----|
| 1. | | 20 |
| A. .. | | 10 |
| | i. Explain the encryption process of the Double Columnar Transposition Cipher with clear steps. Using the plaintext "informationsecurity" and the key "PLANT", demonstrate how the encryption is performed. | 10 |
| <p>----- OR -----</p> | | |
| | ii. Explain the steps involved to convert the plaintext in to cipher text using Playfair cipher. Convert the following Plaintext in to cipher text using Playfair cipher:
Key = THIRDCO , Plaintext = COMPUTER | 10 |
| B. | Describe the working of the AES encryption algorithm with operations performed in each round. | 10 |
| 2. | | 15 |
| A. .. | | 7 |
| | i. Give the steps in RSA algorithm. Perform encryption and decryption using the RSA | 7 |

algorithm for the following data: $p=5$; $q=11$; $e=3$; $M=9$

----- OR -----

ii. Mention the steps involved in the Diffie-Hellman Key Exchange Algorithm. In a Diffie-Hellman key exchange scenario, Alice and Bob agree on a prime number $n=23$ and a base $g=5$. 7

1. Alice selects a private key $x=6$. Compute her public key.

2. Bob selects a private key $y=15$. Compute his public key.

3. Compute the final shared secret key for both Alice and Bob.

B. Draw and explain the internal working of the Feistel Network that forms the backbone of the DES algorithm. For the given data, predict the output after Round 2 in the Feistel network. 8

■ Plaintext: 0011 0101

■ Operation inside the function block: OR

■ Key1 = 0110

■ Key2 = 1001

3. 15

A. What is a buffer overflow vulnerability in computer systems? Explain how it occurs with the help of a simple code example. 7

B. .. 8

i. What is MD5, and how does it transform input data into a fixed-size hash value? 8

----- OR -----

ii. What is HMAC, and how is it different from CMAC? 8

4. 15

7

A. What is Denial of Service (DoS) Attack? Explain Distributed Denial of Service (DDoS) attack in detail and outline the strategies to employ defense against DDoS attack.

B. ..

8

1.

8

What is a firewall? List common types of firewalls and briefly explain any two.

----- **OR** -----

2. What is DNS spoofing in information security? Explain how DNS spoofing attacks are carried out and describe strategies to mitigate such attacks.

8